

MAT 426, Class 2 Homework: Solving Systems with LU Factorization

(Due Thursday, February 12, 2026)

In Exercises 1–5, find an LU factorization of A .

1. $A = \begin{bmatrix} 2 & -4 \\ 3 & 1 \end{bmatrix}$

2. $A = \begin{bmatrix} 2 & 2 & -1 \\ 4 & 0 & 4 \\ 3 & 4 & 4 \end{bmatrix}$

3. $A = \begin{bmatrix} 2 & 2 & 2 & 1 \\ -2 & 4 & -1 & 2 \\ 4 & 4 & 7 & 3 \\ 6 & 9 & 5 & 8 \end{bmatrix}$

4. $A = \begin{bmatrix} 1 & 2 & 0 & -1 & 1 \\ -2 & -7 & 3 & 8 & -2 \\ 1 & 1 & 3 & 5 & 2 \\ 0 & 3 & -3 & -6 & 0 \end{bmatrix}$

5. $A = \begin{bmatrix} 2 & -6 & 6 \\ -4 & 5 & -7 \\ 3 & 5 & -1 \\ -6 & 4 & -8 \\ 8 & -3 & 9 \end{bmatrix}$

6. (a) Suppose A is a nonsingular and has an LU factorization. Prove that this factorization is unique.
- (b) In Example 3 in class, we showed that there were two distinct LU factorizations of the matrix

$$A = \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ -4 & -5 & 3 & -8 & 1 \\ 2 & -5 & -4 & 1 & 8 \\ -6 & 0 & 7 & -3 & 1 \end{bmatrix}$$

Does Example 3 serve as a counterexample to the statement proved in (a)? Explain why or why not.

7. Let $A = \begin{bmatrix} 1 & 2 & 2 \\ 1 & 1 & 1 \\ 2 & 2 & 1 \end{bmatrix}$.

- (a) Find an LU factorization of A and compute LU to show that $LU = A$.
- (b) Reduce A to U using the row operation belows in the order given:

$$R_3 \leftarrow R_3 - 2R_2$$

$$R_2 \leftarrow R_3 - R_1$$

Using this sequence of row operations, find the LU decomposition for A . Then, compute LU and see if $LU = A$. Why might this behavior have occurred?

8.

(a) Let $a_k = \begin{bmatrix} a_{1,k} \\ a_{2,k} \\ \vdots \\ a_{n,k} \end{bmatrix}$. Show that

$$M_k = \begin{bmatrix} 1 & 0 & \dots & \dots & \dots & 0 & 0 & 0 \\ 0 & 1 & 0 & \dots & \dots & \dots & 0 & 0 \\ 0 & 0 & 1 & 0 & \dots & \dots & \dots & 0 \\ \vdots & \vdots & \ddots & \ddots & \ddots & \dots & \dots & \vdots \\ \vdots & \vdots & & 0 & 1 & \ddots & \dots & \vdots \\ 0 & \vdots & & & -\frac{a_{k+1,k}}{a_{k,k}} & \ddots & 0 & \vdots \\ & & & & \vdots & \ddots & \ddots & 0 \\ 0 & 0 & & & -\frac{a_{n,k}}{a_{k,k}} & \dots & 0 & 1 \end{bmatrix}$$

is such that $M_k a_k = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ a_{k,k} \\ 0 \\ \vdots \\ 0 \end{bmatrix}$

- (b) Show that $M_k = I + m e_k^T$ where $m = (0, 0, \dots, 0, -\frac{a_{k+1,k}}{a_{k,k}}, -\frac{a_{k+2,k}}{a_{k,k}}, \dots, -\frac{a_{n,k}}{a_{k,k}})^T$, where e_k is the k -th unit vector.
- (c) Show that the inverse of $M_k^{-1} = I - m e_k^T$.